

FloodLAMP

Biotechnologies, PBC

Delivering the testing our world needs (and people want)
For COVID-19 and beyond

Randy True, Founder and CEO | randy@floodlamp.bio



Bend Fire and Rescue – Site Bring Up

12-01-2021



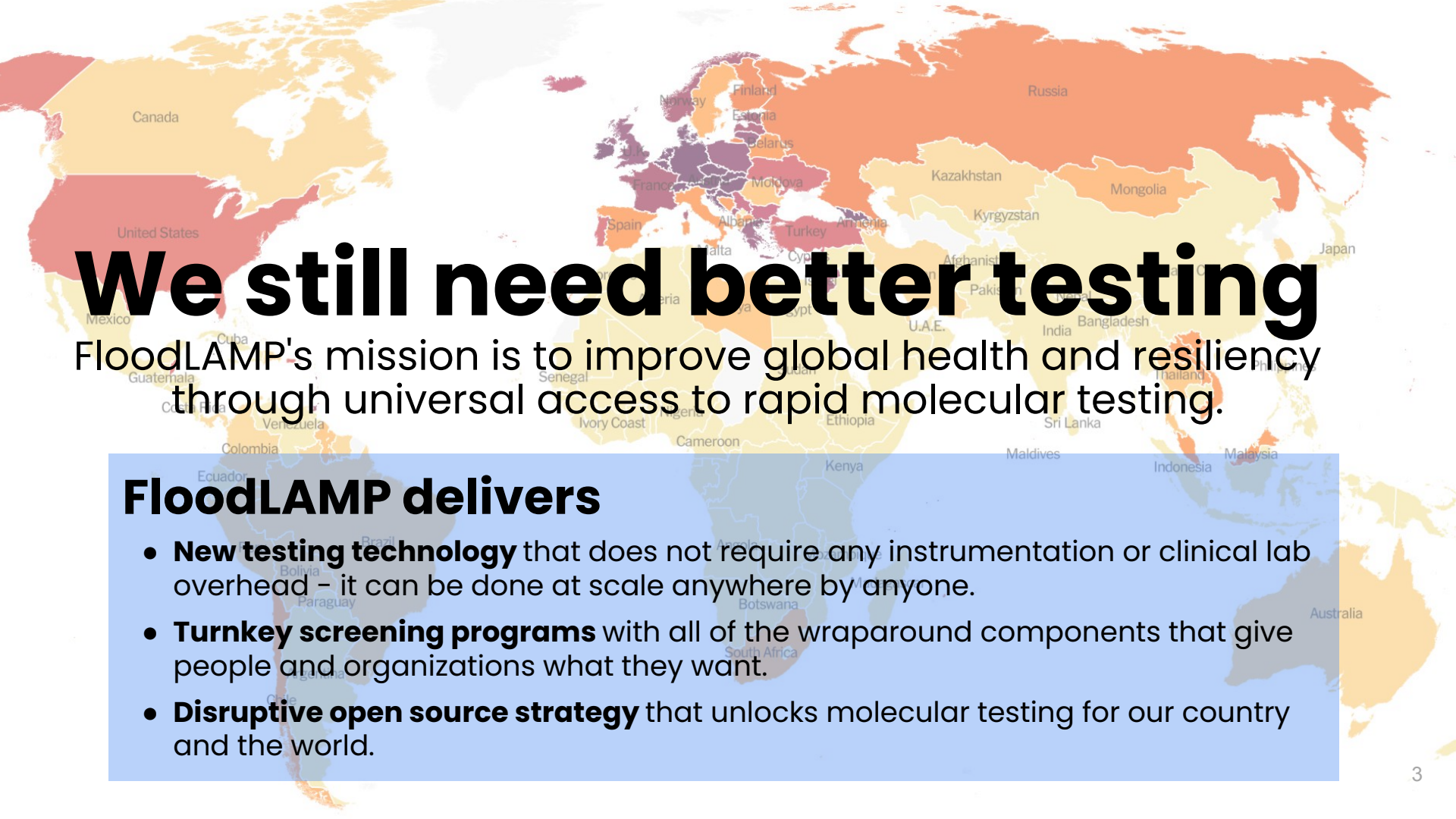
Special Thanks to:

- Katrina Brandis! Volunteer from Bend Science Station
- Kevin Schallert! FloodLAMP COO and Founder NSVD



Today's Agenda

1) Intros	5 min	9:30 am start
2) FloodLAMP Overview	10 min	
3) Safety Highlights	5 min	
4) Contamination Procedures	10 min	10:00 am
5) Test System Overview	10 min	
6) Validation Run	80 min	11:30 am
- Katrina		
7) Training Kit Overview	20 min	
8) Cleanup / Wrapup	10 min	12:00 noon

A world map in the background, color-coded by region: North America (red), South America (orange), Europe (purple), Africa (yellow), Asia (orange), and Australia (yellow).

We still need better testing

FloodLAMP's mission is to improve global health and resiliency through universal access to rapid molecular testing.

FloodLAMP delivers

- **New testing technology** that does not require any instrumentation or clinical lab overhead – it can be done at scale anywhere by anyone.
- **Turnkey screening programs** with all of the wraparound components that give people and organizations what they want.
- **Disruptive open source strategy** that unlocks molecular testing for our country and the world.

Technology Readiness – TRL 6/8



FloodLAMP QuickColor™ COVID-19 Test

Instructions for Use v1.2

IVD

COVID-19 Emergency Use Authorization Only
For in vitro diagnostic (IVD) Use



✓ Successful real world programs:

- With EMS departments in 3 states, screening first responders and city employees;
- Commercial contract screening TV production crew 3X per week with <2hr TAT;
- Local community pre-school testing program with at-home family pooling enabled by the FloodLAMP Mobile App.

✓ Regulatory progress and potential:

- EMS programs green-lit as surveillance by top officials at FDA and CMS;
- 2 full FDA EUA submissions for open source protocol EUA's (direct PCR and LAMP);
- Successful clinical evaluation from Stanford CLIA Lab;
- Pre-EUA for Pooled Swab Collection Kit DTC with FloodLAMP Mobile App;
- IRB approved for clinical study of pooling, home collection and App;
- Experienced consultants and advisors on board, including Anne Wyllie (SalivaDirect).

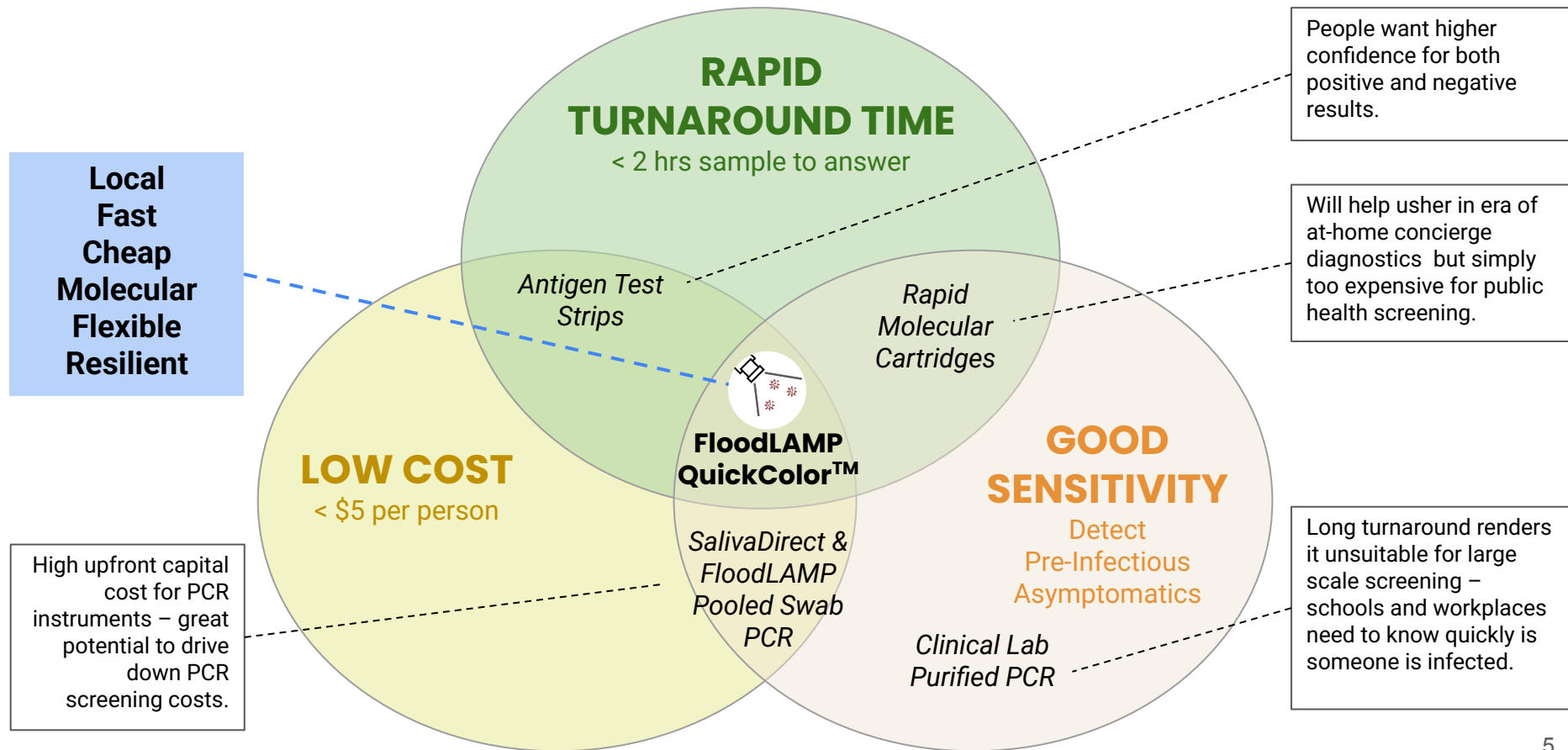
✓ Key IP protection:

- Broad coverage of pooling and digital integration;
- License to core assay chemistry from Harvard Medical School (Rabe-Cepko protocol);
- New game-changing platform IP in development.

✓ Pilot manufacturing:

- Relationships with top reagent suppliers;
- 1.3M reactions of primers ready-to-go in freezers (for 5M people with 4X pooling).

Performance in All Dimensions



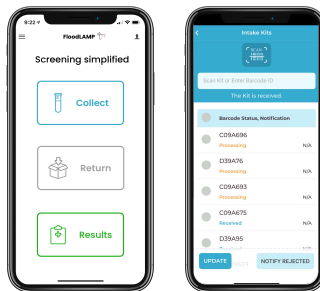
Our Solution

Unique Combination of Advantages in a Fully Integrated Program



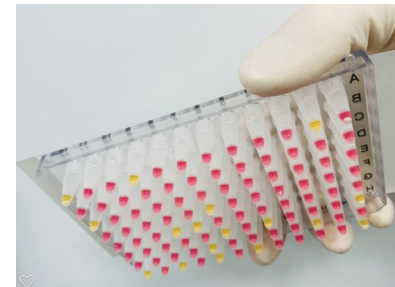
Pooled Swab Collection

- Up to 4 nasal swabs per sample tube



FloodLAMP Mobile App

- Custom app supports both self and sponsored collection
- Results can be reported directly to participants and administrators
- Lab staff interface for batch processing

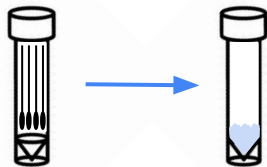


FloodLAMP QuickColor™ Test

- RT-LAMP "PCR-like" molecular amplification
- No instruments required to run test or analyze results
- Low cost and easy to run
- Results in 45 minutes with only 15 minutes hands-on time

Core Assay Technology

FloodLAMP has fully validated 2 complementary tests that are best-of-breed with EUAs submitted to the FDA. Pooled home collection kit was also submitted for interactive review. The test workflow and collection kit has been designed to expand to non-COVID-19 targets in the future.



Streamlined Sample Prep

- Upfront swab pooling
- Highly scalable, integrated processing
- Same sample for both tests



Visual Readout

QuickColor™ LAMP Test

- High sensitivity (90%)
- Ultra-high throughput
- Ideal for serial screening
- Uniquely scales without capital intensive instruments

Licensed from Harvard Medical School (Rabe Cepko Protocol)



Standard RT-qPCR
instrument

EasyPCR™ Test

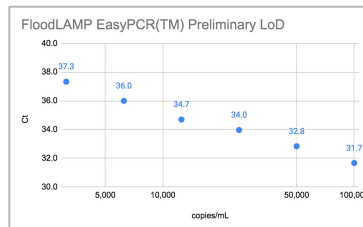
- Very high sensitivity (98%)
- Medium throughput (1.5 hrs/94)
- Ideal for diagnostics/reflex/confirm

Clinical Evaluation Data

Clinical evaluation performed by the Stanford CLIA Lab, with excellent results and praise on the "really straightforward" protocol.

EasyPCR™ Test

- 3 copies/μl LoD
- 98% sensitivity (PPA 39/40)
- 100% Specificity (40/40)
- No false positives



FloodLAMP SwabDirect PCR Result	Comparator Positive	Comparator Negative	Total
Positive	39	0	39
Negative	1	40	41
Invalid	0	0	0
Total	40	40	80
Positive Agreement	97.5% (39/40) 95% CI: 86.8% to 99.9%		
Negative Agreement	100% (40/40) 95% CI: 91.2% to 100%		

QuickColor™ LAMP Test

- 12 copies/μl LoD
- 90% Sensitivity (PPA 36/40)
- Missed positives only high Ct (>36 with direct PCR)
- 100% Specificity (40/40)
- No false positives

FloodLAMP QuickColor Test Result	Comparator Positive	Comparator Negative	Total
Positive	36	0	36
Negative	4	40	44
Total	40	40	80
Positive Agreement	90.0% (36/40) 95% CI: 76.3% to 97.2%		
Negative Agreement	100% (40/40) 95% CI: 91.2% to 100%		

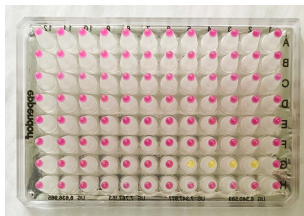
Source of Specimens: Stanford COVID-19 Clinical Testing Program
Specimen Type: Anterior Nares Swab in PBS, previously tested and frozen
Comparator Test: Hologic Panther Fusion SARS-CoV-2 Assay and Hologic Panther Aptima SARS-CoV-2 Assay

Case Studies



Florida Municipal & EMS Deployments

- Davie and Coral Springs FL EMS Depts purchased systems to routinely screen first responders and city staff.
- Full-time light duty firefighters dedicated to running FloodLAMP.
- Unknown positive cases detected and many positives confirmed.
- Clearance from CMS and FDA to operate non-diagnostic "surveillance" testing program
- Reordered test kits and expanding programs.



Teen Summer Camp

- Screened entire camp, 2 cohorts, 300 and 400 campers.
- Remote bringup and processing by local academic volunteers.
- FloodLAMP screening brought up early after 1 positive case identified on Day2.
- Positive case confirmed by FloodLAMP.
- Daily surveillance screened bunkmates and contacts post exposure.



EMS Leadership Conference Screening

- Pop-up lab at conference venue - Hard Rock Hotel Ft. Lauderdale.
- Setup and validation complete in 3 hrs.
- Screened conference attendees on-site in pools of 2-4.
- Same day training of volunteers to run test.
- No positive or invalid results.
- 69 minute average turnaround time in lab



Preschool At-home Family Pooled Collection

- Pilot underway with onboarding and first testing sessions successful.
- Implementing "holy grail" of at-home self-collection of entire families.
- Using FloodLAMP Mobile App for distributed collection and processing.
- Convenient drop-off at school.
- 75min true turnaround-time from dropoff cutoff time to results.

Open EUAs

Path to establishing generics in Dx

	 Typical IVD EUA	 CDC EUA	 SalivaDirect™	 SHIELD	 FloodLAMP EasyPCR™	 FloodLAMP QuickColor™
Disclosure of all chemicals and reagents?	No	Yes	Yes	Yes	Yes	Yes
Chemical and reagents available from multiple vendors?	No	Yes	Yes	No	Yes	Yes, *except NEB LAMP MM
Disclosure of primer sequences?	No	Yes std for PCR	Yes	No ProptryThermo	Yes	Yes
Primers commercially available from multiple vendors?	No	Yes	Yes CDC Primers	No ProptryThermo	Yes CDC SD Primers	Yes Available but not launched
Supply chain robust?	No	No/Maybe	Yes	No/Maybe	Yes	Yes
EUA Sponsor Organization Type	For Profit Company	Govt	Academic Not for Profit	Academic Not/For Profit ?	Public Benefit Corp	Public Benefit Corp
Designation of CLIA labs	Kit Sales	N/A open RoR	Impact & Expansion	Impact & Expansion	Impact & Expansion	Impact & Expansion

Team

Mission driven and execution experience, with a growing network of influential advisors and collaborators.



Randy True
Founder & CEO

- Founder TMI Inc. - Acq. Affymetrix 25MM
- VP of R&D at Affymetrix



Kevin Schallert
COO

- Co-Director covid19sci.org, National Volunteer Scientist DB
- Founder & COO of VineEye



Theresa Ling
UX/Design Lead

Former Uber, New Relic, R/GA



Gary Withey, Ph.D.
Director Process Development

- Current Associate Director, Atreca Inc.
- Former Pilot Production Manager, Affymetrix



Brandon Smith
Lab Assistant

- Silicon Valley CTE Graduate

Scientific Advisory Board

Anne Wyllie

Yale, Lead Researcher – SalivaDirect

Bill Hyun

UC San Francisco, Genoa Ventures

Prof. Connie Cepko (former)

Harvard, Harvard Medical School, Genetics Department, Co-Dir Trans Med Program, HHMI

Industry Advisors

Tim Lugo

William Blair Biotechnology Group Head

Zarak Khurshid

Asymmetry Capital, Top MDx Analyst

John Edge

Oxford Internet Institute, Blue Field Labs, ID2020

Collaborators

gLAMP Group

Global LAMP consortium of 200+ Academic and industry scientists

OpenCOVIDScreen

Jeff Huber (Google, Grail)

Safety Highlights

Protecting You!

Disclaimers:

- This is not a substitute for formal Lab and Biosafety Training.
- Site managers, site personnel, and volunteers are responsible for maintaining appropriate trainings and certifications, and for compliance with local, state, and federal regulations.
- FloodLAMP's safety information is provided on a best-effort basis during this Public Health Emergency.

Resources:

- [UCLA Online Slides/PDFs](#) - [UCLA Root Site](#)
- [Yale - Laboratory Chemical Training by Yale \(40 min\) - free video based](#)
- [Yale - BioSafety Level 2 Training Part1 \(25 min\)](#) [Part2 \(50min\)](#)
- [UC Davis Training ~1 hour online - must request access](#)
- [OSHA Requirements on Hazardous Chemicals in the Workplace](#)

Safety Highlights

PPE



Amp Run Sheet Short Any System

PREP

- ☐ Heaters on (Heat indicator lit!)
- ☐ 1X Inactivation Saline Soln ready
- ☐ Reaction mix strip8/plates ready
- ☐ Safety Procedures:
 - lab coat
 - gloves
 - face mask
 - face shield or goggles
- ☐ Alcohol wipe sample tubes

Safety Highlights

Sample Handling

- Incoming Sample Tubes should be in biohazard bags or in a rack in a closed bin.
- Samples tubes should be wiped with 70% alcohol or ethanol after debuggng.
- Step with the highest risk of infectious exposure is opening the tube with dry swab for addition of the 1X Inactivation Saline Solution (1st step of the Inactivation protocol).
 - recommend performing this step in biosafety cabinet, chem hood, or well ventilated location.



Amp Run Sheet Short Any System

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- ☐ Heaters on (Heat indicator lit!)
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 - lab coat
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 - face shield or goggles
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Safety Highlights

Chem Safety – TCEP Inactivation Solution

FloodLAMP Safety Data Sheets - bit.ly/floodlamp-sds

TCEP - Tris(2-carboxyethyl)phosphine hydrochloride

- Causes severe skin burns / eye damage if contact
- Ingestion / breathing in TCEP is dangerous
- Use in well ventilated area with appropriate PPE

EDTA - Ethylenediamine Tetraacetic Acid

- Causes serious eye irritation / Harmful if inhaled / May cause damage to organs through prolonged or repeated exposure
- Use in well ventilated area with appropriate PPE

	100X IS	1X ISS (Sample)
TCEP	250 mM	2.5 mM
EDTA	100 mM	1 mM
NaOH	1.15 N	11 mM

Give extra care when handling the 100X Inactivation Solution!



Eye Wash Station

- Know location
- Familiarize with use



Sink

- Know location of nearest
- Ensure stocked with soap

Contamination Procedures

Protecting the Tests!

Focus on 4 Types of Contamination

- 1) RNase Contamination
- 2) Positive Control Contamination
- 3) Sample Cross Contamination
- 4) Amplicon Contamination = Death!

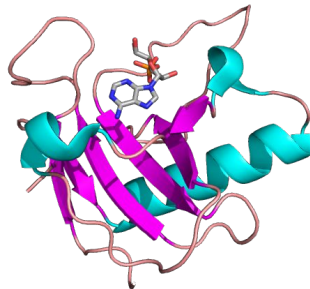
Contamination Procedures

RNase Contamination

<https://www.neb.com/tools-and-resources/usage-guidelines/avoiding-ribonuclease-contamination>

What is RNase and why do we care?

- Enzymes that destroy DNA and RNA, which is what our test detects
- Produced by every living organism in every living cell
- Creates unreliability in test results



When and where do we care?

- Primarily in Amp Reaction

How do we know?

- Positive Controls don't hit (pink or orange)
- BUT this is usually degraded Positive Control - tricky!

RNase is EVERYWHERE!!

What do we do about it?

- **Access Control** - only people who know the rules can come in the room.
- **Gloves, Gloves, Gloves** - don't touch anything except the desk station without gloves.
- **Always Wear Lab Coat** - to avoid forearm slime from getting everywhere.
- **Clean then Keep Clean** - use bins, bags and foil to keep things clean and you life easier.
- **Be Extra Careful w Reaction Plastics** - clean gloves only, don't reach in bags, don't touch inside of caps when opening
- **Be Wary of Cross Contamination**
- **Stage to Replace** - avoid Matlock trap, keep stash clean and secure, and replace if problems

Contamination Procedures

Positive Control Contamination

CDC coronavirus test kits were likely contaminated, federal review confirms

*"It was at this stage of the manufacture, when the bulk reagent materials for the test kits were processed and tested at [the respiratory virus lab], that they were most **likely exposed to positive control material,**"*



How do we know?

- Negative control hit positive (yellow) or inconclusive (orange)
- Suspiciously high number of positive or inconclusive samples

What do we do about it?

- **Follow Protocol** - there are specific procedures and steps explicitly described in the protocol steps and shown in training to take extra care in the handling of the positive controls (TPC and Zeptos).
- Handle tiny TPC tube carefully, open with left hand, keep right hand you pipet with clean.
- Change gloves right after putting TPC back in it's baggie in the freezer.
- Keep positive controls on bottom shelf in freezer, in bags.

Contamination Procedures

Sample Cross Contamination

Means transferring material from one sample tube to another.

For example, a drop from the cap gets on your glove and then you touch the inside of another cap.

How do we know?

- You won't if all samples in batch are negative, which is usually the case!
- Get multiple positives in a batch, especially neighbor tubes. You always confirm positives so if some aren't confirmed but some are, that would indicate sample cross contamination.

What do we do about it?

- **Follow Protocol** - there are specific procedures and steps explicitly described in the protocol steps and shown in training to take extra care to avoid sample cross contamination.
- Do initial wipe of sample tubes with alcohol/ethanol.
- Be careful with caps when opening inactivated samples - we are not spinning them down in a centrifuge so there can be liquid on the cap.
- If you get sample drip/drop, stop and wipe it up carefully, change gloves, clean spot/area with bleach+alcohol. Note sample ID in "notes" on run sheet.

Contamination Procedures

Amplicon Contamination

What are amplicons?

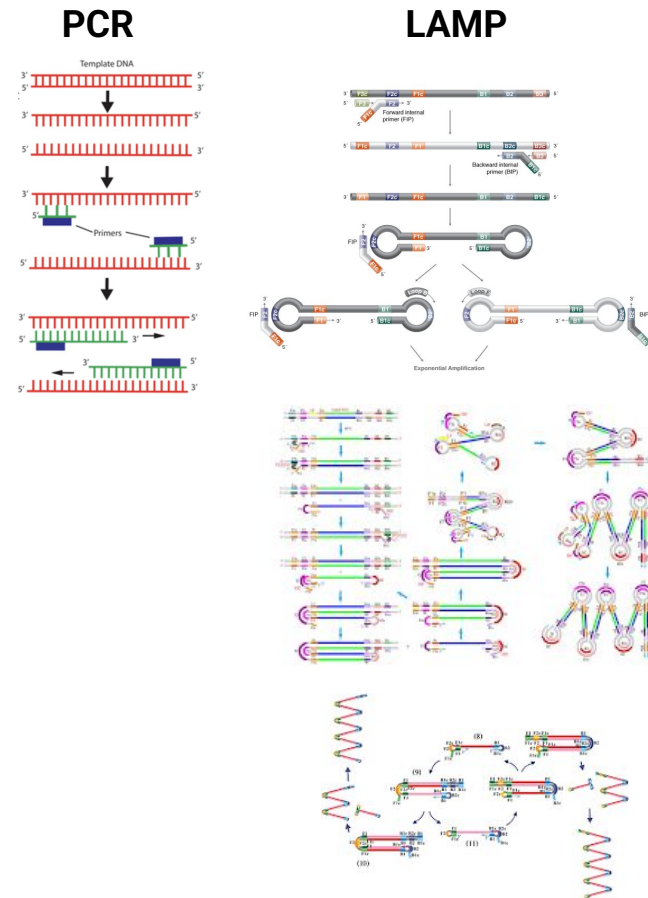
- Amplicons refer to the large amount of DNA produced in a positive amplification reactions. AKA "products".

Why Death?

- Can contaminate entire lab and require start from scratch, sometimes in new building!

How do we avoid Death?

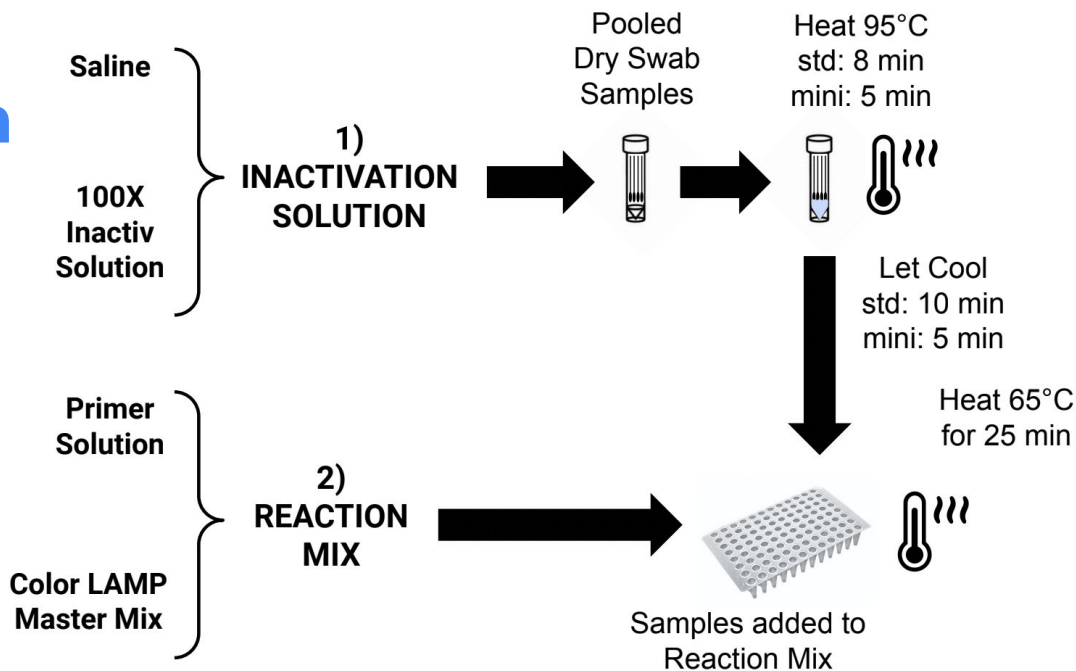
- **NEVER OPEN THE TUBES!** - after amplification.
- Treat the tubes, the heater, the lightbox (anything in the Amp area) as if it's radioactive.
- **Amp Area is a Black Hole** - anything that goes in does not come out: pen, marker, post-it notes.
- **Change Gloves** - after touching amp'd tubes and Amp area.
- Follow these rules and you should be able to avoid amplicon death. Our NEB LAMP Master Mix uses a UDG thermolabile nucleotide (like an extra life, but don't want to use it!).



Test System Overview

2 x 2 x 2:
2 Solution Preps
2 Components Each
2 Heat Steps

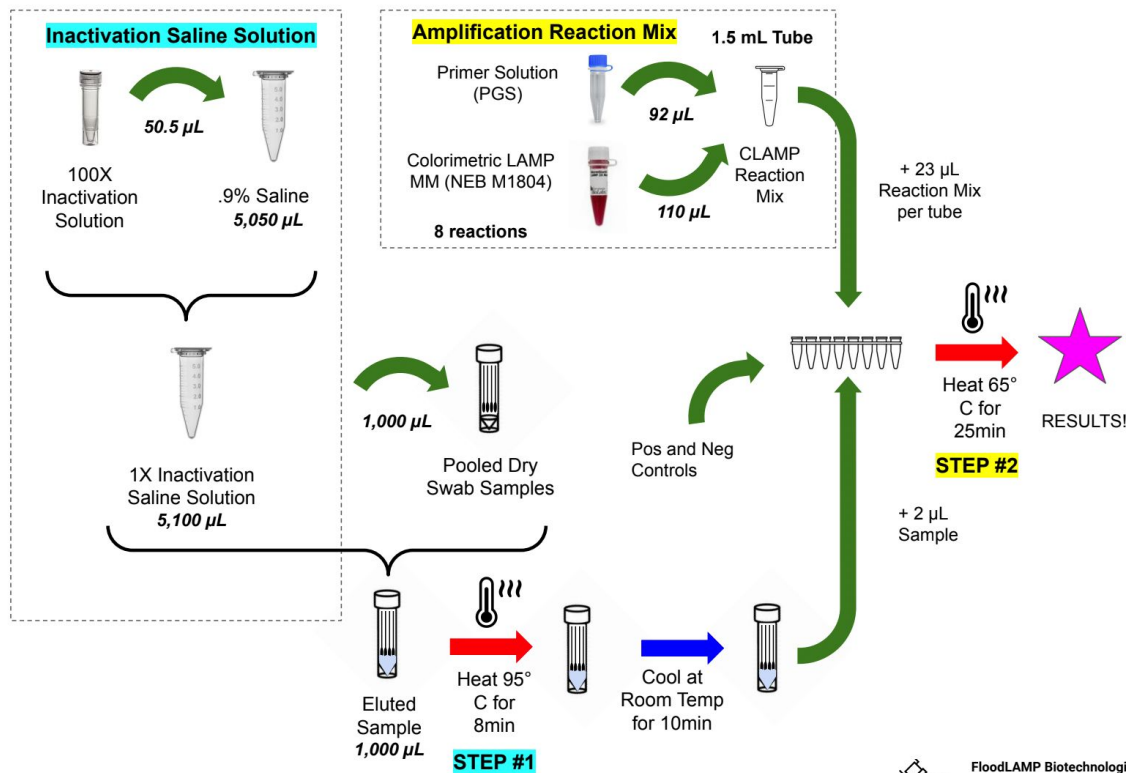
FloodLAMP QuickColor™ COVID-19 Test



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Test System Overview

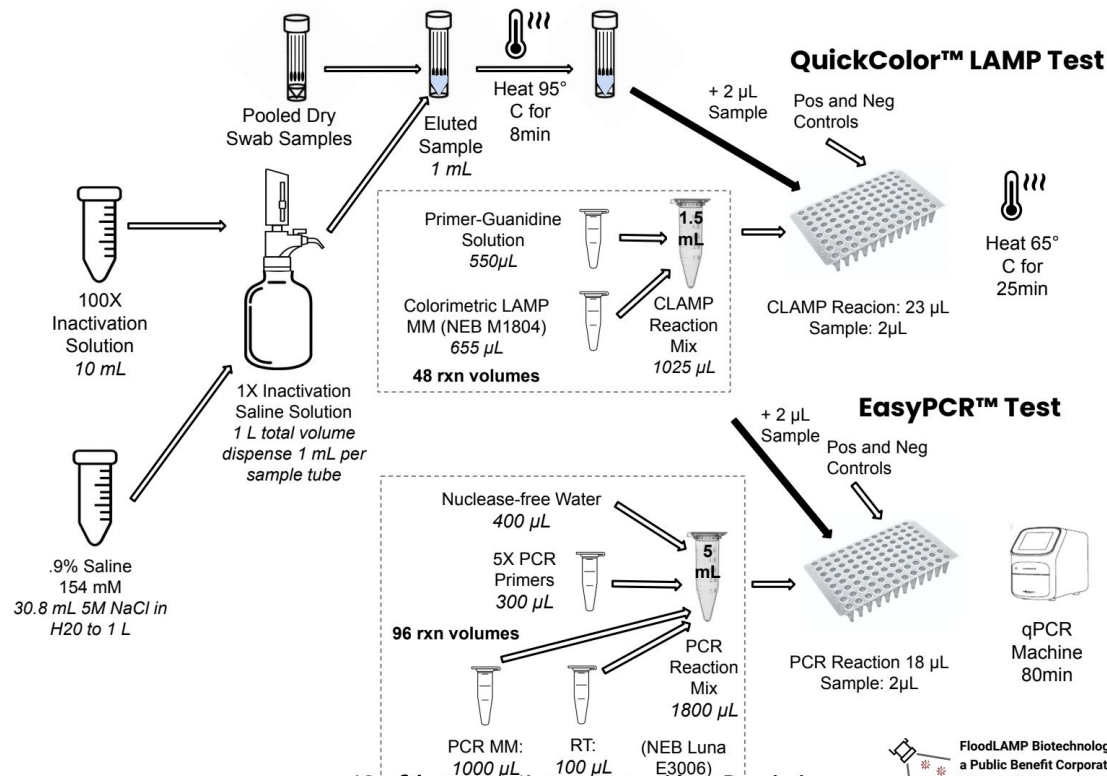
FloodLAMP QuickColor™ COVID-19 Test



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Test System Overview

FloodLAMP Direct RNA Assays



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 FloodLAMP Biotechnologies,
a Public Benefit Corporation
contact@floodlamp.bio

Validation Run

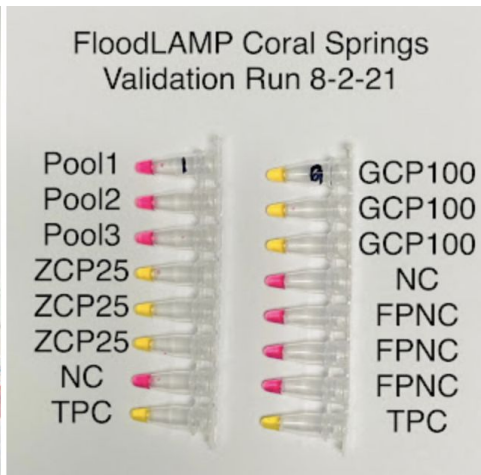
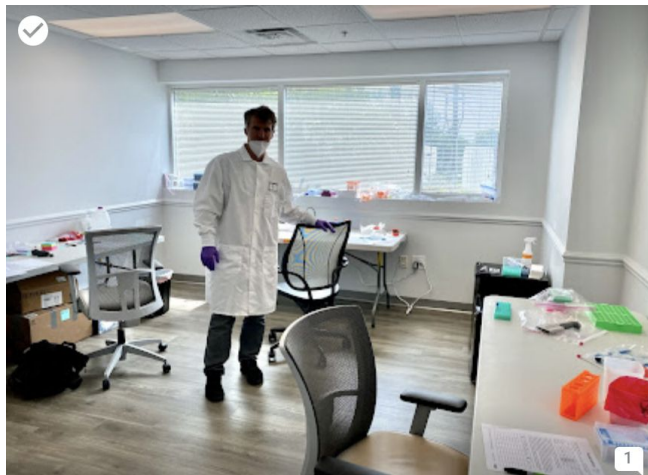
Will create "contrived positive" sample by spiking inactivated virus into the 1XISS
(Zeptos 50uL = 50 cp/uL in ZCP)

Run this in triplicate along with your pooled samples.

Have cold chain concern over reagent shipment - fingers crossed!

Single Strip8:

- 1) TPC
- 2) NC
- 3) ZCP50
- 4) ZCP50
- 5) ZCP50
- 6) Pool1
- 7) Pool2
- 8) Pool3



Training Kit

Learn and practice without burning or contaminating real reagents.

Line #	Item Description	Quantity
1	Training 1.5mL Tubes	30
2	Training 30mL Chub Tubes	2
3	Training 5mL Snap Tubes	2
4	Mock PGS	10
5	Mock CLAMPMM	5
6	Mock 100XIS	1
7	Mock 50mL Saline	2
8	Mock TPC	8